

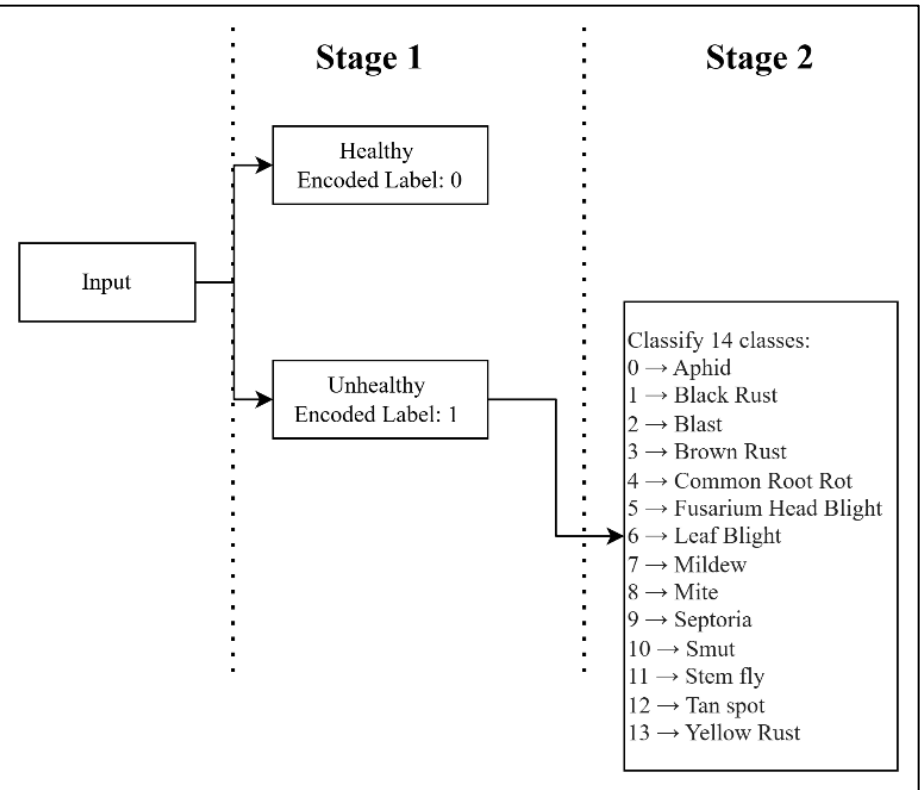
Leveraging AI for Innovative Wheat Disease and Pest Classification and Treatment Solutions

Author: Tam Nguyen

School of Information and Communications Technology - Saskatchewan Polytechnic

INTRODUCTION

- Problem:** In 2024, the Crop Protection Network reported that Alberta and Ontario experienced a **27% loss of wheat** due to diseases, in contrast to just 8.3% in the United States. This underscores the urgent need for early intervention in plant diseases to enhance crop yields and strengthen agricultural economies.
- Objective:** This project aims to use Convolutional Neural Networks to create a two-stage classification model that accurately identifies wheat diseases and pests while offering appropriate treatment options.



APPLICATIONS

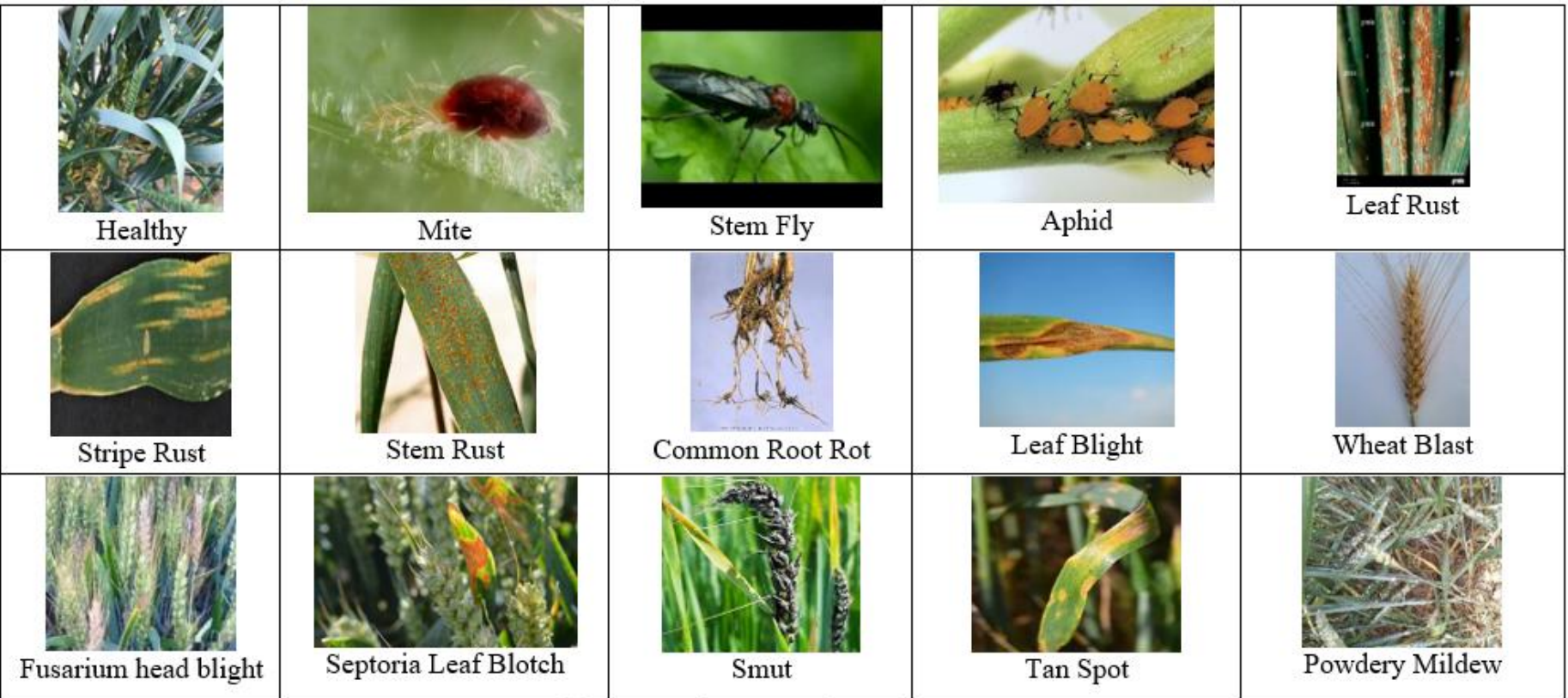
- Real-Time Disease and Pest Detection System for Farmers:** Farmers can take a photo of a wheat leaf using their smartphone, and the system will instantly identify diseases along with specific treatment options.
- Automated Field Monitoring System for Agriculture Companies:** Utilize drones or fixed IoT cameras across wheat farms to continuously capture images, automatically detect diseases and pests, and send alerts to farmers or field officers along with suggestions for immediate treatment actions.
- Decision-Support Tool for Agronomists & Researchers:** The system can assist agricultural scientists in analyzing disease distribution, tracking seasonal outbreaks, evaluating treatment effectiveness, and identifying high-risk regions.
- Web or Mobile Application for Agricultural Education:** A platform for students and farmers to learn about various wheat diseases and pests, including their visual symptoms, characteristics, and suggested treatments.

DATASET EXPLANATION

This dataset contains **13,104 high-resolution images** of real-world wheat diseases, excluding artificially augmented images.

Dataset link: <https://www.kaggle.com/datasets/kushagra3204/wheat-plant-diseases>

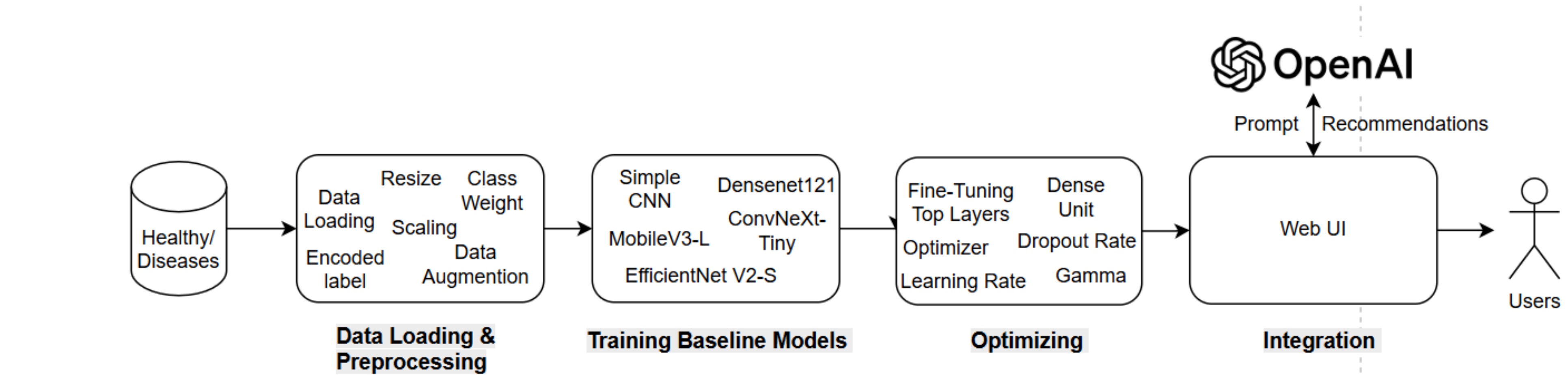
Number of classes: 15 (Healthy, Pest: 3, Leaf/Root diseases: 11)



Sample images from the dataset along with their corresponding ground truth labels.

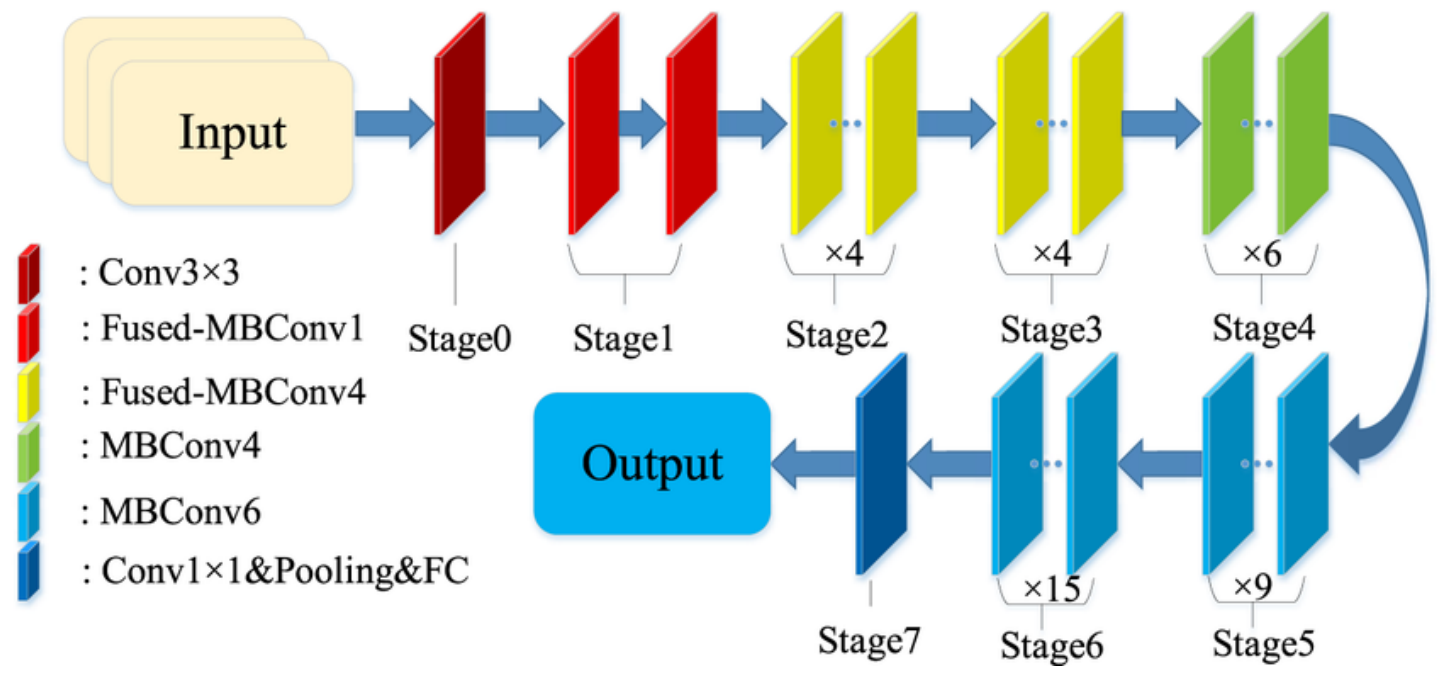
METHODOLOGY

The diagram below shows the main stages of this project.



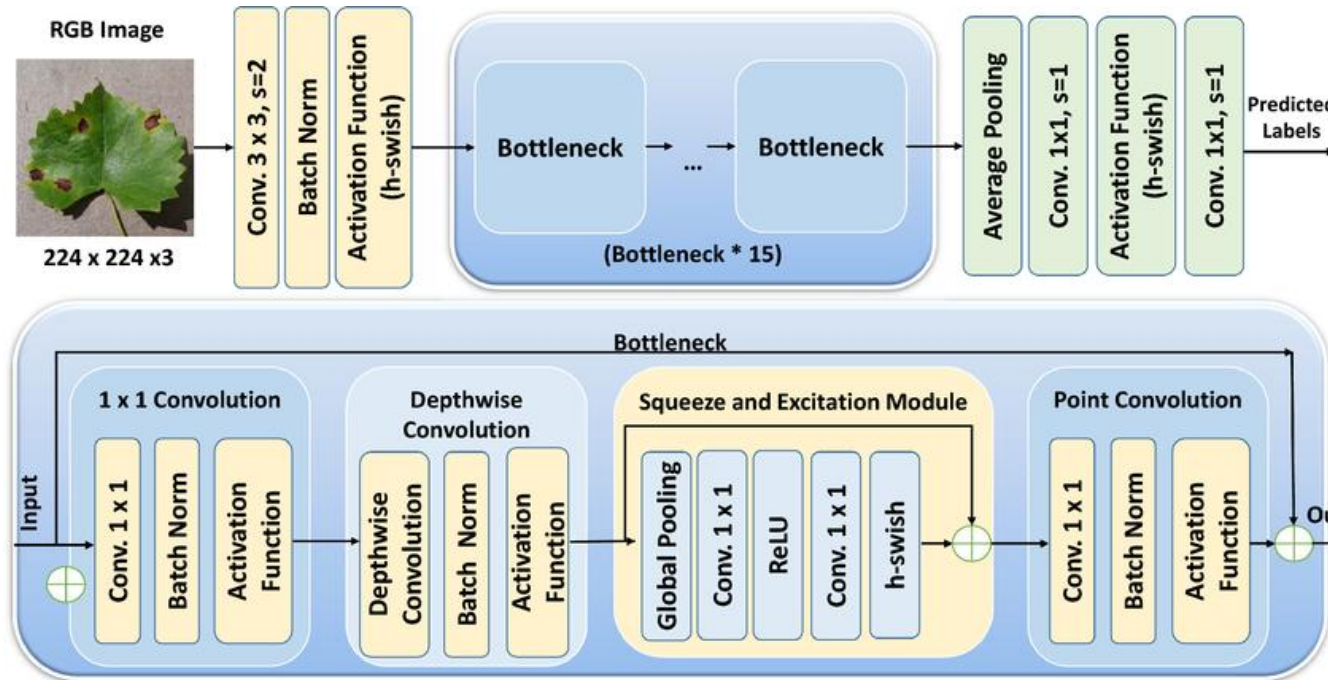
ARCHITECTURE OF PRE-TRAINED MODELS

The EfficientNetV2-S is capable of efficient fine-grained classification, offering a good balance of speed and accuracy. The Fused-MBConv accelerates early feature extraction, boosting efficiency, while deeper MBConv layers with Squeeze-Excitation improve accuracy and capture complex textures.



High-level architecture of EfficientNetV2-S

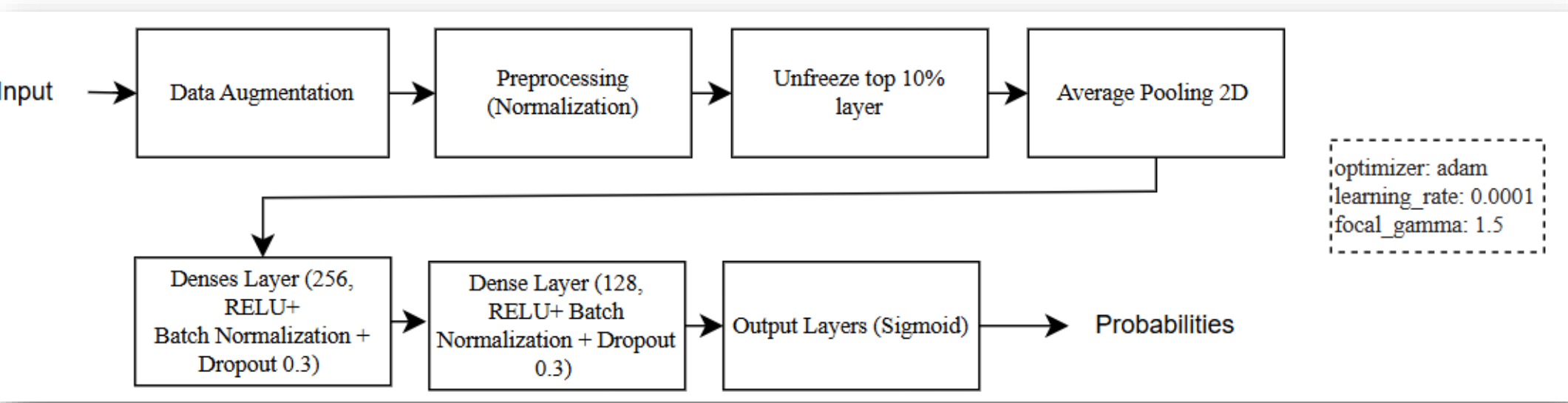
MobileNetV3-Large is a highly efficient, lightweight CNN that balances computational speed with high classification accuracy. It uses the Inverted Residual Block design from MobileNetV2, Hard-Swish activation, and selective SE attention to effectively capture important features with lower computational costs



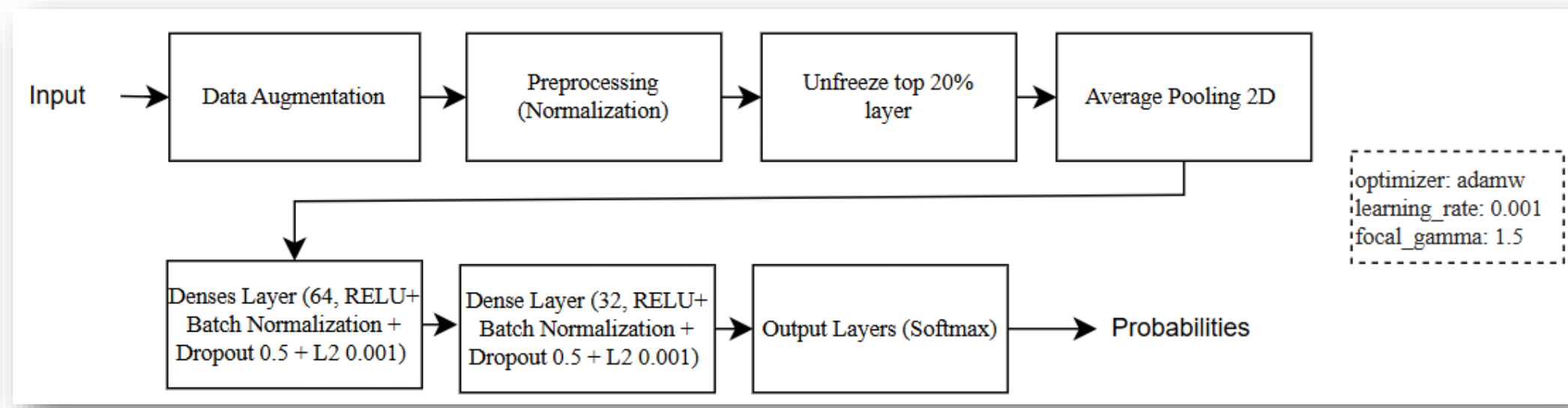
High-level architecture of MobileNetV3-Large

MODEL ARCHITECTURE

Here is the architecture of the final models, which is presented at each stage of classification.



Architecture of the Stage 1 Classification Model



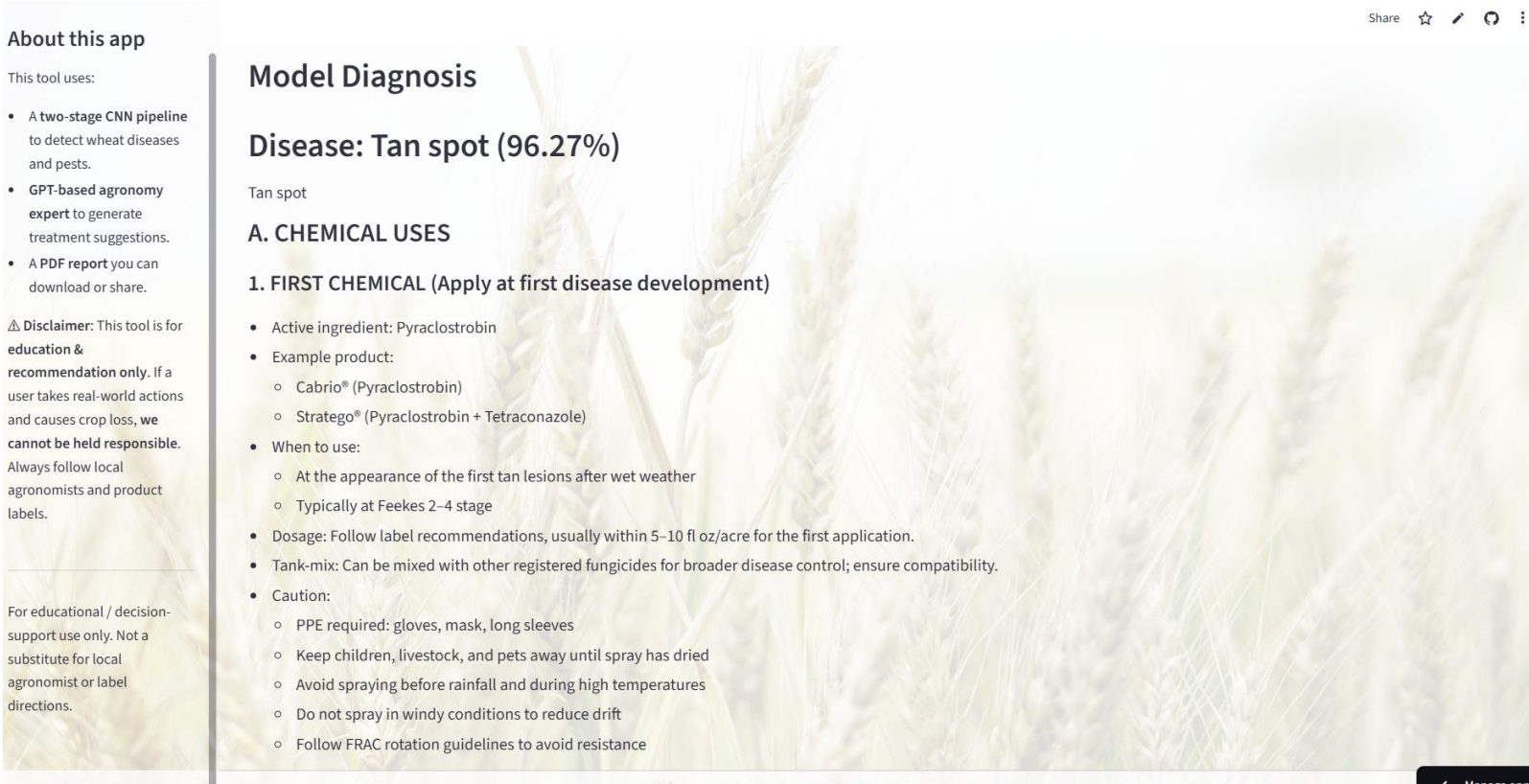
Architecture of the Stage 2 Classification Model

RESULT

After merging two models, the final model achieved **92.17 % accuracy** and a **91.73% F1 score**.



Screenshot of the main page (before uploading the image)



Screenshot of the main page (after uploading the image) - Testing unhealthy case

CONCLUSION

This project aims to develop an AI-powered system that detects diseases and pests in wheat and provides appropriate treatments to enhance wheat yield. Implementing this model in real-world settings brings clear benefits. Reducing crop losses boosts productivity, which drives economic growth by increasing farmers' incomes. Additionally, this system helps farmers cut costs by preventing the use of incorrect treatments. By providing accurate treatment recommendations, we can make agriculture more sustainable.

REFERENCE

- [1] Crop Protection Network. (2024). Wheat disease loss estimates from the United States and Ontario, Canada, 2024. Crop Protection Network. Retrieved from [https://cropprotectionnetwork.org/publications/wheat-disease-loss-estimates-from-the-united-states-and-ontario-canada-2024#:~:text=In%202024%2C%20disease%20reduced%20harvested,vaue%20at%20\\$1.8%20billion%20USD.](https://cropprotectionnetwork.org/publications/wheat-disease-loss-estimates-from-the-united-states-and-ontario-canada-2024#:~:text=In%202024%2C%20disease%20reduced%20harvested,vaue%20at%20$1.8%20billion%20USD.)
- [2] Tan, M., & Le, Q. (2021). Structure of EfficientNetV2-S [Diagram]. Retrieved from https://www.researchgate.net/figure/Structure-of-EfficientNetV2-S_fig3_373371329
- [3] Howard, A., Sandler, M., Chu, G., Chen, L.-C., Tan, M., Wang, W., ... Adam, H. (2019). MobileNetV3-Large architecture [Diagram]. Retrieved from https://www.researchgate.net/figure/MobileNetV3Large-architecture_fig6_382178903